SPVS local ecosystem and GitHub synchronization report

Historical pre-publication audit. The Git gaps, pending file counts and artifact ages below describe the 24 October inspection. They are superseded for publication/delivery status by the 25 October synchronization report. The original findings are retained for traceability.

Audit date: 24 October 2026, Asia/Kolkata. GitHub and local state were checked starting at 19:54 IST; automated evidence was refreshed during this audit. This report covers the local Mac continuation after the last published main commit, including source changes, runtime state, builds, tests, documentation, artifacts, omissions and the remaining synchronization work.

1. Main finding

Local and GitHub are not fully synchronized. The local working branch has four unpublished commits plus further uncommitted changes. GitHub main contains the prior cross-platform implementation, but none of the Mac continuation commits or the newer enrollment/Wi-Fi fixes. The local application source includes the new work and passes the checks recorded below. Physical Android verification, classroom Wi-Fi configuration and production validation remain incomplete.

The local release folder also mixes a new Android APK with older source ZIP, PDF and verification documents. This audit corrects current documentation and the APK checksum, preserves historical evidence, and explicitly identifies the remaining older deliverables. It does not publish, merge or deploy anything.

2. Exact Git position

Repository: Secure Presence Verification System. It is public; default branch is `main`. GitHub metadata/branch/PR inspection and a fresh Git fetch agreed with the local history comparison.

Reference	Observed value
GitHub main / local origin/main	`3591220eb380b7d09a022362a4b9e44c6e684032`
Local main	Same `3591220` commit
Current local branch	`spvs/ios-native-completion`
Current local HEAD	`ac97c60941a13c02b9b197d52fd427adcf219e35`
Divergence from remote main	4 local-only commits; 0 remote-only commits
Remote branch of the current name	Absent
Current branch upstream	Not configured
Staging area at audit start	Empty
Worktree at audit start	21 tracked modified files + 7 untracked files = 28 file entries
Worktree after this documentation audit	39 tracked modified files + 20 untracked files = 59 entries; all unstaged
Tracked file deletions at audit start	None

GitHub main was committed on **23 October at 17:37:47 IST**. At the beginning of this audit it was about **26 hours 17 minutes old**. The useful gap is the four commits and pending file changes, not elapsed time alone. Local main matches GitHub main but is older than the current working branch. Pulling main alone would not upload or checkpoint the local work.

Four local-only commits

Commit	Time (23 October, IST)	Work recorded
`f1b23f5`	19:51:22	Native iOS simulator validation on Apple Silicon, locks, configuration and build workflow
`8f699d9`	23:26:27	iOS Wi-Fi capability becomes opt-in for supported signing teams
`428975e`	23:32:03	Trusted launch on a physical iPhone
`ac97c60`	23:47:29	Physical iPhone login and native device registration

These commits change 18 paths relative to main, with 3,201 insertions and 45 deletions of text lines, largely including the resolved native lockfiles. Those line deletions do not mean entire features or files were removed.

What GitHub currently has

The remote branches are `main`, `spvs/complete-implementation-20261022` at `0ddd78a68d55c37c09704ac9d3880359e07520b2`, and `spvs/final-completion-cross-platform` at `c1e04f09a42c54aa71fc96e68549420c997c7386`. PR #1, PR #2, and PR #3 are merged/closed. No PR exists for this local continuation. The older checkpoint's description of PR #2 as a draft is historical and no longer current.

No GitHub Actions workflow files are tracked in the inspected main/current trees. Local passing tests therefore must not be described as a current GitHub CI run. GitHub is verified here as source hosting. A separate live cloud application or database deployment was not inspected; the old cloud build environment is not evidence of a current production deployment.

3. New work in the local ecosystem

Apple Silicon and native iOS

- Preserved the FastAPI/PostgreSQL, React dashboard and Expo/React Native architecture.

- Installed/resolved CocoaPods and Ruby dependencies and retained `Gemfile.lock`

and `Podfile.lock`. Generated native projects remain excluded except that lock.

- Built the iOS simulator app and the signed physical-device app. The device

launch needed the user's developer-profile trust step before succeeding.

- Confirmed physical iPhone login and signed P-256 device registration. The

client reports Secure Enclave storage; that is not genuine Apple attestation.

- Made Wi-Fi Information an explicit opt-in because the selected Personal Team

could not provision it. Native current-network code remains present.

- Added `--no-clean` to Expo prebuild to preserve native signing settings and locks.

- Removed unused iOS microphone, Face ID, motion and always-location descriptions

from the generated configuration. Camera, foreground location and local-network access remain. These configuration changes did not remove face verification.

See iOS Mac validation for the historical build commands, signing/profile details and the new dated enrollment/attendance evidence.

Enrollment, liveness and retry fixes

- Corrected the unmirrored camera convention: a person's left turn moves their

nose toward image-right. The server now maps that displacement to the person's requested direction.

- Added photo-driven open/closed/reopen blink phases and a vibration to reopen

the eyes. Each new prompt gets 700 ms to settle before capture.

- Moved the instruction above the camera and displays progress through the five

requested poses. Missed-step errors identify the requested turn, center or blink.

- Allows at most two missing-landmark frames, with at least eight tracked frames

still required. A tracking gap resets incomplete pose/blink progress. Every frame still undergoes image quality, single-face detection and identity checks.

- Kept the face-match threshold and challenge/signature/consent requirements.

- Added bounded stage/count diagnostics without retaining raw face frames or

returning biometric vectors/landmark coordinates.

- Added typed API errors, parsing of `Retry-After`, a retry countdown and prevention

of duplicate capture starts. Definitive 4xx errors discard the saved rejected request; network/5xx failures can retain it for idempotent recovery.

- Rate limiting now reports the remaining window rather than the full window on

every rejection. Attempt limits were not removed or increased.

See biometric design and API documentation.

Android Wi-Fi and build tooling

- Replaced the immediate-read scan flow with asynchronous completion handling.

Readings no older than 15 seconds may be reused; otherwise the app waits up to eight seconds for a successful system scan broadcast.

- Shares concurrent scan requests, cleans up listeners/timeouts, and returns an

explicit unavailable/permission-denied result when evidence cannot be obtained.

- Filters stale/future readings, malformed or placeholder BSSIDs, and invalid

signal values. Deduplicates APs and retains up to 50 strongest valid readings.

- Requests coarse and fine location together and requires precise permission.

- Adds **Settings > Wi-Fi setup check** to display actual BSSID/RSSI evidence

locally, without submitting attendance or automatically trusting an AP.

- Added 10 Kotlin lifecycle/filtering tests and additional mobile adapter tests.

- Installed Java 17 and official Android CLI/SDK/native build tools on this Mac.

Gradle's timed-out download was retried through its official URL and checked against the official SHA-256 before use.

- Added an Android build task, automatic detection of the Homebrew JDK/local SDK,

and optional target architecture selection. Corrected the build script's empty array handling for macOS Bash with `set -u`.

- Built and signature-verified a new arm64 development APK. This is not a

standalone production build and has not been installed on an Android phone yet.

See Android Mac validation and the Wi-Fi field procedure.

4. What was removed, replaced or retained

Item	Actual change
Entire tracked source files	None deleted in the local continuation inventory
Immediate Android scan-result read	Replaced by recent-cache reuse or awaited completion
Unconditional iOS Wi-Fi entitlement	Removed from the default Personal Team configuration; supported-team opt-in retained
Unused iOS permission descriptions	Removed as listed above; not a removal of identity/presence verification
Generic rejected-request retry	Replaced by status-aware recovery; rejected challenges cannot be replayed
Failure on any landmark miss	Replaced by bounded recovery with complete sequence and per-frame identity checks
Prior working-copy APK	Replaced by the new native build; the previous bytes remain in Git history
Historical ZIP/PDF/packaged MD files	Preserved; they remain older snapshots
Backend models and face threshold	Retained; no fabricated face scores or weaker matching threshold
Wi-Fi/GPS review policies	Retained; no automatic GPS-only approval enabled
Database schema	No new migration introduced by these local fixes
Historical test evidence	Retained; new evidence has a separate dated directory

5. Real local runtime changes and observations

These are local database/device actions, not changes that a Git push transfers.

- A local PostgreSQL cluster under `.runtime/postgres` contains separate `spvs`

and `spvs_test` databases. Application tests use the guarded test database.

- The API and Metro had stopped during testing; both were restarted, and health,

model readiness and served mobile bundles were verified.

- Physical iPhone device registration is recorded at 23 October 23:45:13 IST.

- The user confirmed successful, easier face enrollment. The database records

`BIOMETRIC_ENROLLED` on **24 October at 14:37:27 IST**.

- The local demo classroom's placeholder location was replaced by the user's

supplied test pin, retaining a 50 m radius. Precise coordinates are deliberately excluded from this shareable report and remain local database configuration.

- Opened a one-hour CS401 test session, ending **24 October at 15:39:40 IST**.

It has expired; the audit observed zero active sessions. A fresh session will be needed for the next attendance test.

- One physical iPhone attendance attempt at **14:43:49 IST** passed identity,

liveness and GPS gates, then returned `PENDING_REVIEW` with reason `IOS_PRESENCE_REVIEW_REQUIRED`. Distance was 3.7 m with about 9.27 m reported horizontal uncertainty.

- That attempt contained `wifi_status=unavailable`, `wifi_source=none`,

`presence_assurance=GPS_ONLY`, and no App Attest assertion. There are **zero official attendance records** for the demo student at this checkpoint.

- The classroom has **zero configured Wi-Fi AP references**. The signed iPhone

app has no Wi-Fi Information or App Attest entitlement. The selected iOS policy requires review even if a current network matches.

- ADB reported **no connected Android device**. No Android scan readings,

phone enrollment, classroom acceptance or outside-room rejection are claimed.

The sanitized runtime evidence contains only the scoped status/timing/decision information. It omits names, student/device IDs, exact coordinates, credentials, keys, templates and frames.

6. Local environment and persistence boundaries

Component	Observed local state
Workspace	`/Users/anishpantham/Downloads/SPVS/SPVS`
Host	Apple Silicon arm64, macOS 27.0 (26A428); user identifies MacBook Air M4
Xcode	27.1 (27A9275)
Node / npm	24.13.0 / 11.6.2
Python	Backend virtual environment, Python 3.12.8
PostgreSQL	Homebrew 16.15, loopback database access
Mobile dependencies	Expo 58.0.17, React Native 0.88.0, React 19.3.0, TypeScript 6.0.3
Ruby / CocoaPods	Previously observed Ruby 3.3.13, Bundler 4.0.23, CocoaPods 1.16.2; locks retained
Java	Homebrew OpenJDK 17.0.21
Android	SDK platform 37.0, build tools 37/36, NDK 27.1.12297006, CMake 3.22.1
Gradle / Kotlin	9.4.1 / observed build plugin 2.2.10
ADB	Platform tools 37.0.1
API / Metro	Both reachable during audit: API health/readiness HTTP 200, models loaded; Metro running
Dashboard	Production bundle freshly built; local port 5173 was not listening at audit

`.runtime`, backend `.venv`, `node_modules`, `work`, generated native projects, native build caches and environment files are deliberately excluded from Git. These may exist locally even when Git reports no change. Local models are also ignored and obtained using checksum-verified download scripts. The old source ZIP includes model assets, but the Git repository does not rely on committing them.

Local environment files, demo passwords, signing identities/provisioning data, private keys, database files and biometric templates must not be added to source control or copied into the report. No such contents were included in this audit. The public repository's absence of these files is intentional, not a sync defect. The service snapshot records the exact probe time and results. Running services can stop when their terminals close.

7. Current validation evidence

Check	Result and boundary
Full backend suite, fresh audit run	120 passed, 0 failures, 0 skipped; includes real-model smoke cases and mocked-vision API contracts
Mobile Vitest, fresh audit run	42 passed, 0 failures; OS/native interfaces are mocked in these tests
Dashboard Vitest, fresh audit run	8 passed, 0 failures
Native Kotlin Wi-Fi suite	10 passed, 0 failures, 0 skipped; callback/filtering logic, not physical radio validation
Total above	180 passing automated tests across these four distinct suites
Mobile TypeScript	Passed during Android implementation validation
Dashboard production build	Passed again during this audit
Android native debug build	Passed in 6m 50s; arm64-v8a; APK signature verification passed
Metro Android/iOS bundle requests	HTTP 200 observed during the local continuation
iOS native builds	Prior local Mac simulator and signed-device builds passed; not rebuilt during this documentation audit
Browser end-to-end	Historical 3 passing cases; not rerun during this audit and not added to the total above
Containers / migration cycle	Historical cloud evidence retained; not rebuilt or rerun as a production deployment in this audit
Physical iPhone	One enrollment success and one pending-review attendance submission observed
Physical Android	Pending

Current JUnit evidence: backend, mobile, dashboard, native Wi-Fi. Earlier focused runs (35/40 mobile or 54/83 backend tests) are superseded for current counts; repeated runs are not additional independent tests.

Deprecation warnings remain in upstream Starlette/httpx, React test renderer, Vite configuration and Gradle/React Native/Expo APIs. A passing build is not a warning-free build. Unit tests and one person's phone success do not establish biometric accuracy, spoof resistance, classroom discrimination or production scale.

8. Dependency security and version freshness

Fresh `npm audit` results are mobile and dashboard.

- Mobile: **5 high-severity affected package entries**, zero critical/moderate/low.

The entries are `@expo/cli`, `@expo/code-signing-certificates`, `expo`, `node-forge` and `zod`. The direct advisory records identify node-forge GHSA-86w9-cpqp-85rv and Zod GHSA-mrhx-c7xc-m3w5. The count includes propagated dependencies, not five distinct root advisories.

- Dashboard: zero findings in the current npm audit.

- Backend: the historical runtime audit reported zero, but `pip-audit` is not

installed in this local virtual environment and was not rerun here. This is explicitly not a current backend vulnerability clearance.

- No package upgrades or forced downgrades were performed during this audit.

npm's proposed Expo 44 downgrade is incompatible with the current app and was not applied. The changed advisory count does not mean this audit patched packages.

"Up to date" here means the state is accurately recorded and checked against GitHub; it does not mean every dependency is the latest release or all security and production work is complete. Compatible dependency remediation needs its own review and validation before distribution.

9. Artifact currency and integrity

Artifact	Current disposition
Android debug APK	Rebuilt locally, 77,979,730 bytes; native scan fix included; arm64 development client
Final source ZIP	39,896,439 bytes, 179 entries, CRC valid; historical source snapshot
Final project PDF	Historical 36-page report; not regenerated from current source
Final setup PDF	Historical 12-page guide; not regenerated from current source
Packaged final MD documents	Historical and unchanged; folder status file identifies their age
FILE_MANIFEST / FINAL_FILE_MANIFEST	Historical source file/hash inventories; regenerate with the next coordinated release
Current ecosystem report + PDF	Newly produced from this audit
Artifact checksum metadata	Corrected for the new APK and annotated with mixed snapshot dates

Current APK SHA-256: `9bd8032bc87b6f74fc86a2fe11897915fcd598081f514473c424294ec54cd3d6`. The old APK hash was `0c3bc2a1d938e0ce49b3e84ec56d4efd4376d8e332e9f1f10859e610cd478e9e`. All other original final-artifact hashes matched their stored records at audit.

The old ZIP lacks the new iOS validation record, Android validation record, Wi-Fi diagnostics, API error helper, CocoaPods lock and Ruby lock. Thus an integrity-valid ZIP is still outdated. It has not been passed off as the source for the newer APK. The packaging selector was corrected to include the tracked `mobile/ios/Podfile.lock` while keeping generated native projects excluded; the next coordinated package must be regenerated from the intended checkpoint.

See release folder status and artifact verification. There is no claimed production-signed APK or packaged IPA. Previous published APK bytes are recoverable from Git; the local working-copy APK was replaced, not preserved at a second historical filename.

10. File-level inventory

The machine-readable repository inventory lists every final modified/untracked path, remote/local SHAs, the four commits, and paths changed in those commits. It is a point-in-time snapshot and will age after subsequent edits. The audit added documentation/evidence beyond the 28 starting entries; those new files are also not yet committed or on GitHub.

Source and test changes pending at audit start

Paths	Purpose
`backend/app/api/verification.py`	Enrollment audit and allowlisted failure details
`backend/app/core/security.py`	Remaining-window Retry-After
`backend/app/services/vision.py`	Direction mapping, bounded landmark recovery, diagnostic frame index
`backend/tests/test_presence_and_liveness.py`, `test_security_workflows.py`, `test_verification_api.py`, `test_vision_models.py`	Updated server regressions and real-model boundary check
`backend/tests/test_capture_recovery.py` (new)	Direction, recovery and identity-check regressions
`mobile/App.tsx`, `src/capture.ts`, `src/protocol.ts`	Capture timing/cues, progress, retry UX and Wi-Fi settings card
`mobile/src/api.ts`, `src/apiErrors.ts` (new helper)	Typed HTTP failures, retry delay and diagnostic propagation
`mobile/src/platform/index.ts`	Android permissions and honest evidence-source status
`mobile/src/WifiDiagnostics.tsx` (new)	Local-only Wi-Fi setup screen
`mobile/modules/spvs-wifi/android/src/main/java/edu/spvs/wifi/SpvsWifiModule.kt`	Android scan bridge and lifecycle
`mobile/modules/spvs-wifi/android/src/main/java/edu/spvs/wifi/WifiScanSession.kt` (new)	Bounded scan coordination and reading filtering
`mobile/modules/spvs-wifi/android/src/test/java/edu/spvs/wifi/WifiScanSessionTest.kt` (new)	Ten native unit tests
`mobile/modules/spvs-wifi/android/build.gradle`	JUnit test dependency
`mobile/tests/api.test.ts` (new), `capture.test.ts`, `platform.test.ts`	HTTP, prompt, permission and scan-source regressions
`.vscode/tasks.json`, `scripts/build_mobile_macos.sh`	Repeatable Android build and local tool discovery
`docs/BIOMETRIC_DESIGN.md`, `docs/TESTING_STRATEGY.md`, `docs/ANDROID_MAC_VALIDATION.md` (new record)	Design, test boundaries and Android build evidence
`release/final/SPVS_ANDROID_DEBUG.apk`	Rebuilt native development binary

Documentation and local corrections made by this audit

- Added this report, its PDF, CURRENT_STATUS.md, dated JSON/XML

evidence and an exact Git inventory.

- Updated README entry points, current test results and known limitations so

historical "no iPhone/Xcode testing" statements no longer describe today's state.

- Added physical enrollment/attendance evidence to the iOS validation record.

- Updated cross-platform design, user manual, API documentation and local setup

guidance for the new capture and Wi-Fi behavior.

- Marked old completion/delivery/checkpoint documents as historical and linked

them to current state, retaining their original run details.

- Corrected APK checksum metadata and added an explicit mixed-age delivery notice.

- Corrected the future source-package selector for the retained CocoaPods lock.

- Fixed the PDF renderer's missing monospace font-family mapping on Macs without

DejaVu fonts; the failure was reproduced while rendering this report, then the actual PDF was regenerated and checked for readable text.

- Did not commit, push, merge, alter verification policy, regenerate historical

ZIP/PDF releases, or copy runtime secrets/database state into the repository.

11. What GitHub is missing

1. The four Mac/iPhone continuation commits and their 18 changed paths.

2. Current face capture, turn-direction, bounded tracking recovery and retry fixes.

3. Current backend diagnostics and the corresponding regressions.

4. Android asynchronous scan handling, native tests, permissions and diagnostics UI.

5. The local Android build task/script refinements and newly built APK.

6. Today's physical iPhone enrollment/attendance evidence and Android build evidence.

7. This audit, current status/index, updated documentation, fresh test/audit results

and corrected artifact metadata.

GitHub should **not** receive the local database, biometric templates, secret environment files, signing keys, generated native build trees or dependency caches. No live cloud database synchronization is implied by bringing source branches up to date.

12. Completion and publication checklist

Work item	State / next action
Local source and unit/build checks	Current work is present and the recorded checks pass
Local documentation	Updated by this audit; use the current index rather than historical final reports
Git checkpoint	Pending: review and commit tracked modifications plus all intended new source/docs/evidence files
Remote publication	Pending: push `spvs/ios-native-completion` with upstream, then open a reviewed PR into main
Main synchronization	Pending: merge only after review; then refresh the local main branch
Coordinated release package	Pending: regenerate source ZIP/PDF reports from a named checkpoint and verify checksums together
Android physical test	Connect phone, install APK, sign in and run Wi-Fi setup check; no scan results yet
Classroom Wi-Fi references	Confirm actual institution APs and collect inside/corridor/adjacent-room RSSI before configuring thresholds
Next attendance session	Previous demo session expired; create a fresh bounded test session when the phone is ready
Android full attendance	Use a separate test student or the explicit audited device reset flow to preserve one-device binding
iPhone network capability	Supported provisioning and actual network references are still required; paid membership does not provide general scans
Automatic iPhone Wi-Fi approval	No such mode is implemented; needs a separately reviewed policy/design with real evidence
Genuine provider attestation	Apple App Attest and Play Integrity still need configuration and physical/provider verification
Production readiness	Calibration, spoof evaluation, reliable classroom discrimination, compatible security updates, TLS, operational controls and load testing remain

A future publication can use `git push -u origin spvs/ios-native-completion` **after** the new work is committed. Pushing now would publish only the four existing commits, not the uncommitted fixes or this report. Do not force-push. No publication was performed as part of this documentation request.

13. Evidence interpretation

GitHub SHA/branch/PR state came from live GitHub tools and a fresh fetch. Local file and commit state came from Git. Test counts came from actual JUnit results; build outcomes came from actual tool execution. Enrollment/attendance status came from a sanitized read-only local database query and the user's physical-phone confirmation. Historical build details remain explicitly attributed to their dated validation documents. Missing tests, deployments and provider integrations are labelled unverified rather than inferred from successful compilation.